

Given:

$$f(x) = \begin{cases} \sin(2x) & \text{for } x > 0 \\ a + bx & \text{for } x \leq 0 \end{cases}$$

the condition is that ① $f'(0^-)$ and $f'(0^+)$ are equal

② $f(x)$ is continuous at $x=0$

from ②

$$\sin(0) = a + b(0)$$

$$\Rightarrow \boxed{a = 0}$$

from ①

$$f'(0^-) \Rightarrow 2\cos(2x) \Big|_{x=0^-} \\ \Rightarrow 2(1)$$

from ①

$$f'(0^+) = b \\ \Rightarrow b$$

$$\text{So, } \boxed{2 = b}$$

$$\therefore a = 0 \text{ and } b = 2.$$